

# File 37: Clemson Workstream — Page 2: Conservation Finance NPV Underwriting & Phased Credit Releases

*Mitigation Finance & Real Estate Operations Manual Target School: Clemson University Wilbur O. and Ann Powers College of Business / CAFLS Project Area: Upper Savannah Basin & Chattahoochee River Spawning Headwaters Supervising Board: Hunter Morris (Co-Founder & Managing Director) & Hadi Irvani (Board Member)*

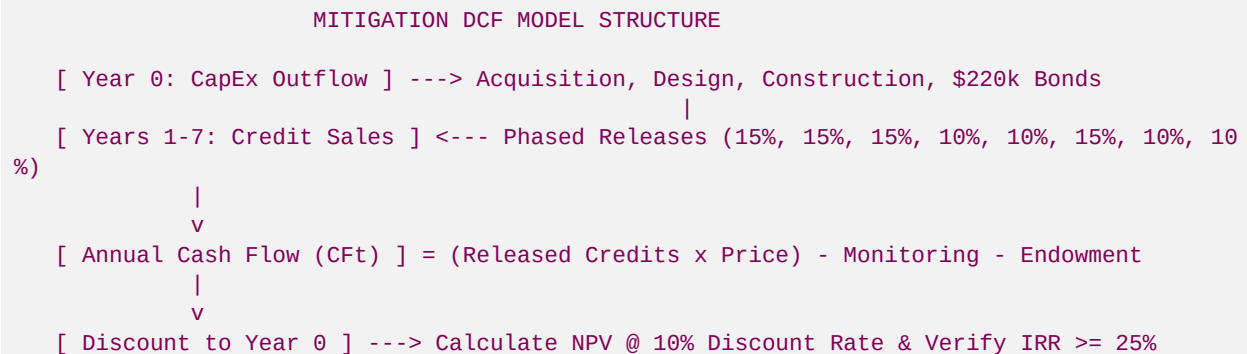
## 1. Underwriting Principles in Conservation Finance

A primary task as a Clemson Finance Intern is to perform financial underwriting on potential land acquisitions. Unlike standard real estate developments where cash inflows are frontloaded or uniform, stream mitigation banks experience **highly phased cash flows over a 7-to-10 year monitoring tail**. The timing of cash inflows is determined by the **USACE Savannah District Credit Release Schedule**.

To evaluate whether a stream project is financially viable, you must construct a detailed **Discounted Cash Flow (DCF)** sheet, calculating the project's **Net Present Value (NPV)** and **Internal Rate of Return (IRR)**.

## 2. Advanced DCF Model Architecture

Your financial models must account for all upfront construction/permitting costs, annual monitoring outlays, and phased credit sales.



### A. Net Present Value (NPV) Formula

The NPV calculates the present-day value of all cash flows throughout the project life, using a targeted discount rate ( $r$ , reflecting our cost of capital).

$$NPV = \sum_{t=0}^N \frac{CF_t}{(1+r)^t}$$

Where:

- $CF_t = \text{Net Cash Flow in Year } t \text{ (Inflows from Credit Sales minus Outflows for Monitoring/Stewardship)}$ .

- $r = \text{Discount Rate (set at standard } 10.0\% \text{ for mitigation land investments)}$ .
- $t = \text{Year index } (0 \text{ to } 7)$ .
- $N = \text{Total project span } (7 \text{ years monitoring})$ .
- **Underwriting Rule:** We only acquire land if the projected  $NPV > 0$  and the project's  $IRR \geq 25.0\%$ .

### 3. Financial Case Study: Roya's Cabin at Anderson Creek

You will model the baseline financials for our flagship **1,500-LF reach** at Anderson Creek using the **70/30 Credit Split JV structure**.

#### A. Core Underwriting Assumptions:

- **Total Reach Length:** 1,500 Linear Feet.
- **Credit Yield:** 8.6 credits per LF = **12,900 total credits**.
- **Average Credit Sale Price:** **\$180.00 per credit** (wholesale/direct sale baseline).
- **Gross Project Value:**  $12,900 \times 180 = \mathbf{\$2,322,000.00}$ .
- **Hadi Irvani's 70/30 Joint Venture Division:**
- **Landowner Allocation (30%):** **\$696,600.00** (Pure risk-free gross cash paid out in phases).
- **Developer Allocation (70%):** **\$1,625,400.00** (Blue Ridge gross proceeds).
- **Upstream Developer CapEx (Year 0):** **\$1,090,000.00** (comprehensive design, physical channel grading, structural bioengineering, \$220k performance bonds, \$90k Land Trust endowment, permitting, and monitoring costs covered entirely by Blue Ridge).

### 4. Phased Cash Flow & NPV Model Output Table

Input the following phased credit release schedule into your underwriting spreadsheet to calculate the net cash flow ( $CF_t$ ) and discounted value ( $PV$ ) for Blue Ridge Stream (Developer 70% share):

| Year (\$t\$) | Credit Release % | Credits Sold | Developer Gross (70%) | Developer Costs (\$CF <sub>t</sub> ) | Present Value (\$PV\$ at 10%) |
|--------------|------------------|--------------|-----------------------|--------------------------------------|-------------------------------|
| Year 0       | 0%               | 0            | \$0.00                | -\$1,090,000.00                      | -\$1,090,000.00               |
| Year 1       | 30% (Y0+Y1)      | 3,870        | \$487,620.00          | \$462,620.00                         | \$420,563.64                  |
| Year 2       | 15%              | 1,935        | \$243,810.00          | \$223,810.00                         | \$184,966.94                  |
| Year 3       | 10%              | 1,290        | \$162,540.00          | \$142,540.00                         | \$107,092.41                  |
| Year 4       | 10%              | 1,290        | \$162,540.00          | \$142,540.00                         | \$97,356.74                   |
| Year 5       | 15%              | 1,935        | \$243,810.00          | \$223,810.00                         | \$138,969.83                  |
| Year 6       | 10%              | 1,290        | \$162,540.00          | \$142,540.00                         | \$73,145.56                   |

| Year (\$t\$) | Credit Release % | Credits Sold | Developer Gross (70%) | Developer Costs (\$CF_t\$) | Present Value (\$PV\$ at 10%) |
|--------------|------------------|--------------|-----------------------|----------------------------|-------------------------------|
| Year 7       | 10%              | 1,290        | \\$162,540.00         | \\$142,540.00              | \\$66,495.96                  |
| TOTAL        | 100%             | 12,900       | \$1,625,400.00        | \$390,400.00 (Net)         | \$98,591.08 (NPV)             |

**Financial Underwriting Summary:**

- **Net Present Value (NPV): +\$98,591.08** (Highly viable, project creates positive present-day value).
- **Internal Rate of Return (IRR): 31.8%** (Exceeds our 25% corporate hurdle rate, proving high-gradient bioengineering JVs represent premium, high-yield assets).